

KIERAN PEREIRA

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EDUCATION

University of California, Berkeley, USA**August 2024 – May 2025***MEng Mechanical Engineering, Concentration in Robotics***Relevant Coursework:** Dynamics & Control of Autonomous Systems, Robotic Locomotion, Design of Microprocessor Systems.**Capstone Project:** *Developing robotic ocean sensors for swarm coordination.*

- Designed and integrated hardware, software and firmware components for a fleet of 100+ autonomous robotic units.
- Evaluated several closed-loop control strategies (PID, LQR) achieving a 15% improvement in steady state performance.
- Conducted ocean field testing with 18 units, validating mesh network communication, autonomous control strategies, and hardware performance for robust real-world operation.

University College London (UCL), London, UK**September 2020 – 2024***BEng Mechanical Engineering with Control for Intelligent Systems – First Class Honors (4.0 GPA)*

- Capstone Project (4.0 GPA): Created an integrated car dashboard system that recognized and narrated the meaning of traffic signs for inexperienced drivers. Leveraged a custom neural network and refined training processes on 100,000 images using PyTorch. Achieved state-of-the-art precision, outperforming Meta's benchmark model.

PROFESSIONAL EXPERIENCE

Undaunted Robotics, Head of Robotics – Founding Engineer, Atlanta, USA**July 2025 – Present**

- Grew and led robotics as the fleet scaled from 10 to 150 robots in the field (largest robotic security provider in the Southeast). Owned the hardware, embedded software, reliability and deployment work.
- Built a security rover as the lead of a 3-person team. Field deployed in 2 months (10 currently active). Architected and implemented embedded software for motor control and teleoperation. Built relationships with local and international manufacturers to build DFM processes to scale towards 200 units / yr.
- Redesigned the robot charging hub, unblocking ~\$620K in stalled deal flow while taking heat-related fleet outages to zero, sustained 12 months.
- Diagnosed a single-vendor dependency after a cloud outage took entire fleet robot offline, architected control independent of the vendor's cloud and shipped it as a 5-min-per-robot field upgrade in 1 week.
- Replaced off-the-shelf camera with a custom vision stack, improving reliability and unlocking new deal flow through a proprietary customer feed and police-integration capability, driven directly by sales and customer feedback.

Lockheed Martin, Software & Systems Engineer, UK**June 2023 – June 2024**

- Led an object detection initiative to improve QA processes, collaborating with an international team to present engineering and financial plans to senior executives, securing \$104k for a pilot.
- Delivered high-quality code in an embedded firmware environment, reviewing production code to quality standards.
- Collaborated directly with customers and manufacturers to formulate engineering requirements and user features for safety-critical systems, architecting these into 600+ design requirements for engineering teams.

University College London, Machine Learning Researcher, London, UK**June – September 2022**

- Developed real-time image classification for skin disease diagnostics, achieving 90% accuracy using a phone camera.
- Collaborated closely with clinicians in the NHS to access private clinical datasets, implementing preprocessing steps to improve performance by 23%.
- Implemented signal processing and feature extraction techniques to improve model performance by 15%.

LEADERSHIP EXPERIENCE

CTO, Obsidian- Full Motion Tracking Wearables, Berkeley, USA**August 2024 – May 2025**

- Founded and led an interdisciplinary team of 7 to design and develop a connected wearable system that enhances athletic performance across multiple sports through real-time motion tracking and personalized feedback.
- Integrated IoT-enabled hardware and sensors, a machine learning recommendation system and sensor fusion algorithms (Kalman Filters) achieving 95% precision to actual movement.
- Implemented state estimation algorithms through various sensor readings (IMU, Strain gauges, barometer).

Lead Controls Engineer, IMechE UAV Challenge, London– 4th / 32**September 2022 – May 2023**

- Led a team in the design and implementation of an image recognition guidance system for an autonomous helicopter, architecting robotic perception and motion planning algorithms in C++.
- Coordinated cross-functional collaboration across 6 teams (42 engineers), ensuring we met 256 design requirements.

RELEVANT SKILLS & INTERESTS

Programming: C++, Python, ROS2, Git, I2C, Linux, Test Automation, Embedded Firmware, MATLAB, Simulink, Data Analytics**Robotics:** Sensor design & fusion, Motor Controller Operation, Controls, PCB Design, State Estimation, Kalman Filters**Non-Technical:** Technical Communication, Stakeholder Management, Agile, Data Analysis, Product Management, Mentorship